

Mock Midterm 2 — B

Worked solutions and marking guide

18 points: Exercise 1 (8), Exercise 2 (10)

These are full solutions, not only final answers. The marking allocations are for self-assessment, not an official course rubric. Any equivalent network construction or correct derivation can receive full credit. Give partial credit for a correct method even if a later arithmetic error propagates.

Exercise 1: A nonlinear Bayes rule and a small neural network

1a Posterior log odds and a bent boundary

[2 p]

Bayes' rule cancels the evidence:

$$a(x) = \log \frac{\pi_1}{\pi_2} + \log \frac{p(x_1 | C_1)}{p(x_1 | C_2)} + \log \frac{p(x_2 | C_1)}{p(x_2 | C_2)}.$$

The prior term is $-\log 2$. The Laplace ratio contributes

$$\log 2 - |x_1| + \frac{1}{2}|x_1| = \log 2 - \frac{1}{2}|x_1|.$$

The Gaussian normalizers cancel, and their log ratio is

$$-\frac{1}{2} \left[(x_2 - \frac{1}{2})^2 - (x_2 + \frac{1}{2})^2 \right] = x_2.$$

All constants therefore cancel:

$$a(x) = x_2 - \frac{1}{2}|x_1|, \quad p(C_1 | x) = \sigma(x_2 - \frac{1}{2}|x_1|).$$

The strict decision region is

$$R_1 = \{x : x_2 > \frac{1}{2}|x_1|\}.$$

The boundary is a V with vertex at the origin: the ray $x_2 = x_1/2$ for $x_1 \geq 0$ and the ray $x_2 = -x_1/2$ for $x_1 \leq 0$. The region is above both rays. On the boundary, both posteriors are $1/2$.

Naive Bayes says that the log odds are a sum of one-coordinate functions. It does *not* require each function to be linear in its coordinate. The Laplace log ratio introduces an absolute value, giving a piecewise-linear rather than a globally affine decision rule.

Suggested credit: 1 p for the simplified posterior/log odds; 0.5 p for the region and boundary; 0.5 p for the distinction between conditional independence and linearity.

1b Correct labels, nonlinear features, and nonexistence of a finite MLE [2 p]

Let an affine logit be $g(x) = w_0 + w_1x_1 + w_2x_2$, with $g > 0$ for C_1 . Because $(0, 1)$ is the midpoint of the other two points,

$$g(0, 1) = \frac{1}{2}[g(-4, 1) + g(4, 1)].$$

If both outer points have strictly negative logits, their midpoint does too. This contradicts the required positive logit at the middle point. No affine classifier can strictly classify the three points.

One valid fixed feature map and weight vector are

$$\phi(x) = (1, |x_1|, x_2)^\top, \quad w = (0, -\frac{1}{2}, 1)^\top.$$

They give $w^\top \phi(x) = a(x)$ for every input, not only the three points. The three logits are $+1, -1, -1$, so the transformed data are strictly linearly separable. Multiplying w by $q > 0$ preserves all predicted labels and increases their correct-class probabilities. The negative log-likelihood is

$$L(qw) = 3 \log(1 + e^{-q}) \longrightarrow 0 \quad \text{as } q \rightarrow \infty.$$

At any finite weight vector, each sigmoid probability lies strictly between zero and one, so every correctly labeled observation has a strictly positive loss. The likelihood supremum is 1 but is not attained at finite weights:

There is no finite unregularized MLE for this separable dataset.

Finite weights can give perfect classification accuracy without assigning probability 1 to the observed labels. A positive definite Gaussian weight prior would prevent this unlimited scaling for logistic regression with fixed features.

Suggested credit: 0.75 p for the affine impossibility proof; 0.5 p for exact nonlinear features; 0.75 p for the finite-MLE distinction and scaling argument.

1c An exact two-unit construction with a direct connection

[1 p]

Use

$$U = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad c = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad d = 0.$$

Then $h_1 = \text{ReLU}(x_1)$ and $h_2 = \text{ReLU}(-x_1)$, and

$$c^\top h + v^\top x + d = -\frac{1}{2}[\text{ReLU}(x_1) + \text{ReLU}(-x_1)] + x_2 = x_2 - \frac{1}{2}|x_1|.$$

The hidden units form the absolute-value term. The direct connection supplies x_2 without using extra hidden units. All parameters are finite. Hidden-unit permutations and positive rescalings of ReLU units give equivalent constructions.

Suggested credit: 0.75 p for a valid exact construction; 0.25 p for identifying the two paths.

1d Backpropagation through both paths

[2 p]

First differentiate the binary cross entropy and sigmoid:

$$\frac{\partial L}{\partial y} = -\frac{t}{y} + \frac{1-t}{1-y}, \quad \frac{dy}{da} = y(1-y), \quad \boxed{\delta := \frac{\partial L}{\partial a} = y - t}.$$

Let $I_i = \mathbf{1}\{r_i > 0\}$ and let $D_h = \text{diag}(I_1, I_2)$. From $r_i = \sum_j U_{ij}x_j + b_i$ and $a = \sum_i c_i h_i + \sum_j v_j x_j + d$, component derivatives are

$$\begin{aligned} \frac{\partial L}{\partial U_{ij}} &= \delta c_i I_i x_j, & \frac{\partial L}{\partial b_i} &= \delta c_i I_i, \\ \frac{\partial L}{\partial c_i} &= \delta h_i, & \frac{\partial L}{\partial v_j} &= \delta x_j, & \frac{\partial L}{\partial d} &= \delta, \\ \frac{\partial L}{\partial x_j} &= \delta \left(\sum_i c_i I_i U_{ij} + v_j \right). \end{aligned}$$

Assembled gradients and the numerator-layout input derivative are

$$\begin{aligned} \nabla_U L &= \delta (D_h c) x^\top \in \mathbb{R}^{2 \times 2}, & \nabla_b L &= \delta D_h c \in \mathbb{R}^2, \\ \nabla_c L &= \delta h \in \mathbb{R}^2, & \nabla_v L &= \delta x \in \mathbb{R}^2, \\ \boxed{\frac{dL}{dx}} &= \delta (c^\top D_h U + v^\top) \in \mathbb{R}^{1 \times 2}. \end{aligned}$$

The input derivative is the *sum* of the hidden-layer path and the direct path; omitting $v^\top$ gives the wrong result.

For the specified numerical check,

$$r = (1, -2)^\top, \quad h = (1, 0)^\top, \quad a = 0, \quad y = \frac{1}{2}, \quad \delta = -\frac{1}{2}, \quad D_h c = (1, 0)^\top.$$

Therefore

$$\nabla_U L = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \frac{dL}{dx} = (-\frac{1}{2}, -1).$$

For reference, the other gradients are $\nabla_b L = (-\frac{1}{2}, 0)^\top$, $\nabla_c L = (-\frac{1}{2}, 0)^\top$, $\nabla_v L = (-1, 0)^\top$, and $\partial L / \partial d = -\frac{1}{2}$.

Suggested credit: 0.25 p for the logit derivative; 1.25 p for component and assembled derivatives, including both input paths; 0.5 p for the two numerical results.

1e Zero initialization cannot learn the hidden features

[1 p]

At initialization $U = b = c = 0$, so $r = h = 0$ for every training input. Under $\text{ReLU}'(0) = 0$, $D_h = 0$. The formulas in 1d give

$$\nabla_U L = 0, \quad \nabla_b L = 0, \quad \nabla_c L = \delta h = 0$$

for every observation, regardless of the current v, d . An isotropic Gaussian prior adds gradients proportional to the parameters, which are also zero for U, b, c . A gradient step therefore leaves all three at zero. This repeats at every subsequent step, proving the claim by induction.

The resulting predictor is always $\sigma(v^\top x + d)$, an affine-logit classifier. By 1b it cannot strictly classify all three points correctly. Weight decay cannot move a parameter away from zero when both its data gradient and its penalty gradient are zero. A nonzero, appropriately scaled hidden initialization avoids this particular frozen state; this is an optimization failure, not a lack of representational capacity of the full network.

Suggested credit: 0.5 p for the gradient/induction argument; 0.5 p for the capacity reduction and why the prior cannot revive the layer.

Exercise 2: Soft routing among regression experts

2a Joint, marginal, and posterior expert probabilities

[2 p]

Write $m_{nk} = \theta_k^\top \phi_n$ and $f_{nk} = \mathcal{N}(t_n | m_{nk}, s^2)$. With one-hot expert indicators,

$$p(t, Z | X, \Theta, V, s^2) = \prod_{n=1}^N \prod_{k=1}^K [\pi_{nk} f_{nk}]^{z_{nk}}.$$

For each observation exactly one expert is active. Summing over its possible values gives

$$p(t | X, \Theta, V, s^2) = \prod_n \left(\sum_k \pi_{nk} f_{nk} \right), \quad \ell = \sum_n \log \left(\sum_k \pi_{nk} f_{nk} \right).$$

By Bayes' rule,

$$r_{nk} = \frac{\pi_{nk} f_{nk}}{\sum_j \pi_{nj} f_{nj}}, \quad \sum_k r_{nk} = 1.$$

The experts are alternative explanations for one target, not independent observations of that target. Their densities must be added with routing weights. Multiplying them neither marginalizes a categorical latent variable nor generally gives the stated normalized mixture density.

Suggested credit: 0.5 p each for complete likelihood, observed likelihood/log-likelihood, responsibilities, and the sum-versus-product explanation.

2b Observed-likelihood gradients and an unidentifiable gate

[2 p]

Let $p_n = \sum_j \pi_{nj} f_{nj}$. For an expert parameter component,

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_{kr}} &= \sum_n \frac{\pi_{nk} f_{nk}}{p_n} \frac{\partial \log f_{nk}}{\partial \theta_{kr}} \\ &= \sum_n r_{nk} \frac{t_n - \theta_k^\top \phi_n}{s^2} \phi_{nr}. \end{aligned}$$

Thus the column gradient is

$$\nabla_{\theta_k} \ell = \frac{1}{s^2} \sum_n r_{nk} (t_n - \theta_k^\top \phi_n) \phi_n \in \mathbb{R}^M.$$

For the routing parameters, differentiate the softmax in components:

$$\frac{\partial \pi_{nj}}{\partial v_{kr}} = \pi_{nj} (\delta_{jk} - \pi_{nk}) \phi_{nr}.$$

Consequently

$$\begin{aligned} \frac{\partial \ell}{\partial v_{kr}} &= \sum_n \frac{1}{p_n} \sum_j f_{nj} \pi_{nj} (\delta_{jk} - \pi_{nk}) \phi_{nr} \\ &= \sum_n \left(r_{nk} - \pi_{nk} \sum_j r_{nj} \right) \phi_{nr} = \sum_n (r_{nk} - \pi_{nk}) \phi_{nr}, \end{aligned}$$

so

$$\nabla_{v_k} \ell = \sum_n (r_{nk} - \pi_{nk}) \phi_n \in \mathbb{R}^M.$$

These formulas come directly from differentiating the log of a sum. Although the responsibilities depend on current parameters, they appear as the result of the derivative; there is no additional $\partial r / \partial \theta$ term in these final formulas.

If all expert densities are the same, say $f_{nk} = f_n$, then $p_n = f_n \sum_k \pi_{nk} = f_n$ and $r_{nk} = \pi_{nk}$. Hence every routing gradient is zero and the likelihood is independent of V . The data provide no information about routing between identical experts. This is a stronger ambiguity than the usual softmax common-shift invariance.

Suggested credit: 0.75 p for each gradient with a component derivation; 0.5 p for the identical-expert conclusion.

2c EM updates and the condition for monotonicity

[3 p]

Initialize finite expert and routing parameters and $s^2 > 0$, preferably using different initial experts. At iteration j :

E-step. Use the old parameters to calculate

$$r_{nk}^{(j)} = \frac{\pi_{nk}^{(j)} \mathcal{N}(t_n \mid (\theta_k^{(j)})^\top \phi_n, (s^2)^{(j)})}{\sum_\ell \pi_{n\ell}^{(j)} \mathcal{N}(t_n \mid (\theta_\ell^{(j)})^\top \phi_n, (s^2)^{(j)})}.$$

M-step or generalized M-step. Hold these responsibilities fixed in

$$Q(\Theta, V, s^2) = \sum_{n,k} r_{nk}^{(j)} \left[\log \pi_{nk}(V) - \frac{1}{2} \log(2\pi s^2) - \frac{(t_n - \theta_k^\top \phi_n)^2}{2s^2} \right].$$

Do not recompute or differentiate the responsibilities inside this objective. The old posterior defines the expectation; the new parameters are its optimization variables.

Let $\Phi \in \mathbb{R}^{N \times M}$ have rows $\phi_n^\top$, $t \in \mathbb{R}^N$, and $D_k = \text{diag}(r_{1k}^{(j)}, \dots, r_{Nk}^{(j)})$. Differentiation of Q with respect to an expert component gives

$$\frac{\partial Q}{\partial \theta_{kr}} = \frac{1}{s^2} \sum_n r_{nk}^{(j)} (t_n - \theta_k^\top \phi_n) \phi_{nr}.$$

Setting this to zero yields the weighted normal equations and update

$$\Phi^\top D_k \Phi \theta_k = \Phi^\top D_k t, \quad \boxed{\theta_k^{(j+1)} = (\Phi^\top D_k \Phi)^{-1} \Phi^\top D_k t}.$$

The inverse exists by assumption. This weighted least-squares update is independent of the shared variance. The factor $1/s^2$ cancels, but the responsibilities still depended on the old variance in the E-step.

Set $A = \sum_{n,k} r_{nk}^{(j)} [t_n - (\theta_k^{(j+1)})^\top \phi_n]^2$. Because $\sum_{n,k} r_{nk}^{(j)} = N$, the variance-dependent part of Q is $-\frac{N}{2} \log s^2 - A/(2s^2)$. Treating s^2 as the optimization variable,

$$\frac{\partial Q}{\partial s^2} = -\frac{N}{2s^2} + \frac{A}{2(s^2)^2} = 0, \quad \boxed{(s^2)^{(j+1)} = \frac{1}{N} \sum_{n,k} r_{nk}^{(j)} [t_n - (\theta_k^{(j+1)})^\top \phi_n]^2}.$$

The denominator is N , not NK , since the responsibilities of each observation sum to one. Positivity of A gives an interior variance maximum.

The routing objective is $Q_V = \sum_{n,k} r_{nk}^{(j)} \log \pi_{nk}(V)$, a soft-target multinomial logistic log-likelihood. Its gradient is

$$\nabla_{v_k} Q_V = \sum_n (r_{nk}^{(j)} - \pi_{nk}(V)) \phi_n.$$

One valid generalized-EM update starts at the old V and uses

$$v_k^{\text{new}} = v_k^{\text{old}} + \eta \sum_n (r_{nk}^{(j)} - \pi_{nk}(V^{\text{old}})) \phi_n,$$

with backtracking on $\eta > 0$ until $Q_V(V^{\text{new}}) \geq Q_V(V^{\text{old}})$. Further steps may be taken with the *same fixed* responsibilities. A gradient direction alone is not a guarantee: an excessive step can reduce Q_V . Increasing the frozen-responsibility Q gives the standard EM lower-bound argument for a nondecreasing observed-data likelihood. Exact expert/variance maximization and a verified nondecreasing routing block suffice.

After the updates, evaluate the new observed-data log-likelihood. For example, stop if its nonnegative relative improvement is less than ε ,

$$\frac{\ell^{(j+1)} - \ell^{(j)}}{1 + |\ell^{(j)}|} < \varepsilon,$$

subject to a maximum iteration count and a numerical check that the likelihood did not decrease beyond round-off tolerance. A parameter-change or gradient check can also be used. Multiple initializations reduce sensitivity to local solutions. EM/GEM does not guarantee a global optimum of this nonconvex observed-likelihood problem; monotonicity is not global optimality.

Suggested credit: 0.5 p for the E-step and frozen-responsibility Q ; 0.75 p each for expert and variance updates; 1 p for valid routing ascent, a convergence check, and the qualified monotonicity/global-optimum statements.

2d Routing and responsibility can select different experts

[1 p]

Before seeing the target, hard routing chooses expert 1 because $4/5 > 1/5$. After seeing $t = 2$, the two Gaussian densities are

$$f_1 = \frac{1}{\sqrt{2\pi}} e^{-9/2}, \quad f_2 = \frac{1}{\sqrt{2\pi}}.$$

Thus

$$r_1 = \frac{4e^{-9/2}}{1 + 4e^{-9/2}} \approx 0.04255, \quad r_2 = \frac{1}{1 + 4e^{-9/2}} \approx 0.95745.$$

The highest-responsibility choice is now expert 2. Routing uses only the input. Responsibility also uses how well each expert explains the observed target; expert 2's much larger density outweighs its smaller routing probability.

Suggested credit: 0.25 p for hard routing; 0.5 p for responsibilities and posterior selection; 0.25 p for the explanation.

2e Predictive uncertainty and squared-error prediction

[2 p]

At a new input x , let $\pi_k = \pi_k(x)$ and $m_k = \theta_k^\top \phi(x)$. By conditioning on the latent expert,

$$\bar{m}(x) = \mathbb{E}[T \mid x] = \sum_k \pi_k m_k.$$

Each expert has second moment $s^2 + m_k^2$. Therefore

$$\begin{aligned} \text{Var}(T \mid x) &= \sum_k \pi_k (s^2 + m_k^2) - \bar{m}^2 \\ &= \boxed{s^2 + \sum_k \pi_k (m_k - \bar{m})^2}. \end{aligned}$$

The first term is within-expert noise. The second is between-expert uncertainty from not knowing which expert generates the target. With expert-specific variances the first term would instead be $\sum_k \pi_k s_k^2$; here it is simply s^2 .

For any scalar prediction $q(x)$,

$$\mathbb{E}[(T - q)^2 \mid x] = \text{Var}(T \mid x) + (\bar{m} - q)^2,$$

because the cross term involving $T - \bar{m}$ has zero conditional expectation. Thus the optimal squared-error prediction is $q(x) = \bar{m}(x)$, not necessarily the mean of the most likely expert.

In the numerical case,

$$\bar{m} = \frac{4}{5}(-1) + \frac{1}{5}(2) = -\frac{2}{5} = -0.4,$$

$$\text{Var}(T \mid x) = 1 + \frac{4}{5}\left(-\frac{3}{5}\right)^2 + \frac{1}{5}\left(\frac{12}{5}\right)^2 = \frac{61}{25} = 2.44.$$

Responsibilities $p(z = k \mid x, t)$ require the very target that is being predicted. Substituting the values from 2d would use target information that is unavailable before observing T . For an unobserved target, marginalize the expert using $p(z = k \mid x)$, the routing probabilities.

Suggested credit: 0.5 p for mean and variance; 0.5 p for the squared-error argument; 0.5 p for the numerical values; 0.5 p for distinguishing routing from target-conditioned responsibilities.

Scope and source alignment

The format follows `exams/homework-based-exam-1.pdf`: two extended exercises, two hours, 18 points, derivations, and follow-up insights. The actual second midterm's question count and topic selection are unknown.

- Exercise 1 develops HW 2 Exercises 1–2, practice sets 3–4, and lectures 6–8. New elements are Laplace/Gaussian class conditionals, a V-shaped Bayes boundary, a new three-point obstruction, exact ReLU features with a direct connection, derivatives through two input paths, and frozen hidden initialization.
- Exercise 2 develops HW 2 Exercise 6, practice set 5, and lecture 9. It replaces the exponential experts with Gaussian regression experts and adds weighted normal equations, a learned shared variance, a verified generalized-EM routing step, indistinguishable experts, and predictive variance.
- Together with Mock A, the pair covers all six HW 2 themes: Naive Bayes, classifier capacity, regularized logistic regression, PCA, generalized Gaussian latent coordinates in PPCA, and mixtures of experts.

The foundational operations necessarily recur, but no original homework or practice-set problem, dataset, or complete derivation sequence is reproduced. The questions are intentionally harder than the first midterm; the point total does not imply equal workload.